

Feynman integral $12e|2e|e|:nnn$

1 Summary

Loops	1
Propagators	3
External legs	3
Monomials of F	6
Monomials of G	9
Independent invariants	6
Codimension	5
Scaleless	no
Polytope vertices	6
Normalised volume	7
Polytope automorphisms	6
Toric generators	14
Landau surfaces	14

2 Graph

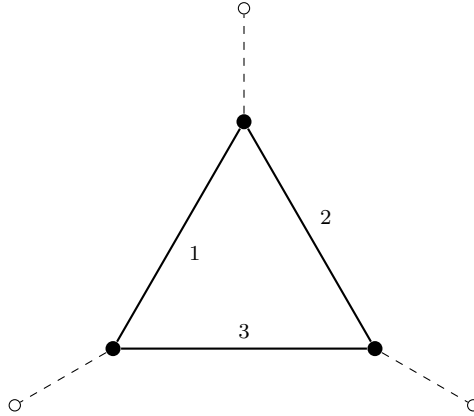


Figure 1: Feynman diagram topology. Internal vertices are shown as filled circles, internal propagators as solid lines, and external legs as dashed lines.

Figure 1 shows the graph. It has $L = 1$ loop, $N = 3$ propagators and 3 external legs.

e	Vertices	ν_e	m_e
1	1-2	ν_1	m_1
2	1-3	ν_2	m_2
3	2-3	ν_3	m_3

Propagator e joins the vertices listed and carries the exponent ν_e and the mass m_e shown in the table; massless propagators have $m_e = 0$. The figure labels each propagator with its index e , which also names its parameters a_e and u_e .

3 Conventions

The integral is

$$I = e^{L\epsilon\gamma_E} (\mu^2)^{\nu-LD/2} \int \prod_{r=1}^L \frac{d^D k_r}{i\pi^{D/2}} \prod_e \frac{1}{(-q_e^2 + m_e^2)^{\nu_e}}$$

in D dimensions [1], with loop momenta k_r , the momentum q_e of propagator e fixed by momentum conservation at the vertices, the metric signature $(+, -, \dots, -)$ and $\nu = \sum_e \nu_e$, here $\nu = \nu_1 + \nu_2 + \nu_3$. Dimensional regularisation sets $D = D_0 - 2\epsilon$ for an even integer D_0 chosen when the result is expanded. The energy scale μ makes I and the Symanzik polynomials dimensionless. External momenta are incoming, and the kinematics is written in the invariants p_1^2, p_2^2 and p_3^2 : the external masses p_i^2 . The second Symanzik polynomial is $F = -\sum_{T_2} s_{T_2} \prod_{e \notin T_2} a_e + U \sum_e m_e^2 a_e$, divided by μ^2 , the sum running over spanning two-forests T_2 and s_{T_2} being the square of the momentum flowing from one tree of T_2 to the other [1].

4 Symanzik polynomials

In the Schwinger parameters a_e the first Symanzik polynomial, the sum over spanning trees T of $\prod_{e \notin T} a_e$, is

$$U = a_1 + a_2 + a_3$$

and the second is

$$\begin{aligned} F = \frac{1}{\mu^2} & (a_1^2 m_1^2 + a_1 a_2 m_1^2 + a_1 a_2 m_2^2 \\ & - a_1 a_2 p_1^2 + a_1 a_3 m_1^2 + a_1 a_3 m_3^2 \\ & - a_1 a_3 p_2^2 + a_2^2 m_2^2 + a_2 a_3 m_2^2 \\ & + a_2 a_3 m_3^2 - a_2 a_3 p_3^2 + a_3^2 m_3^2) \end{aligned}$$

In the Lee-Pomeransky parameters u_e their sum is

$$\begin{aligned} G = U + F = & u_1 + u_2 + u_3 \\ & + \frac{1}{\mu^2} (m_1^2 u_1^2 + m_1^2 u_1 u_2 + m_1^2 u_1 u_3 \\ & + m_2^2 u_1 u_2 + m_2^2 u_2^2 + m_2^2 u_2 u_3 \\ & + m_3^2 u_1 u_3 + m_3^2 u_2 u_3 + m_3^2 u_3^2 \\ & - p_1^2 u_1 u_2 - p_2^2 u_1 u_3 - p_3^2 u_2 u_3) \end{aligned}$$

U has degree $L = 1$ and F degree $L + 1 = 2$; F has 6 monomials and G has 9. The coefficients of G are the GKZ variables z_j , here at their physical values:

j	Monomial	z_j
1	u_1^2	$\frac{m_1^2}{\mu^2}$
2	$u_1 u_2$	$\frac{m_1^2}{\mu^2} + \frac{m_2^2}{\mu^2} - \frac{p_1^2}{\mu^2}$
3	$u_1 u_3$	$\frac{m_1^2}{\mu^2} + \frac{m_3^2}{\mu^2} - \frac{p_2^2}{\mu^2}$
4	u_2^2	$\frac{m_2^2}{\mu^2}$
5	$u_2 u_3$	$\frac{m_2^2}{\mu^2} + \frac{m_3^2}{\mu^2} - \frac{p_3^2}{\mu^2}$
6	u_3^2	$\frac{m_3^2}{\mu^2}$
7	u_1	1

j	Monomial	z_j
8	u_2	1
9	u_3	1

The configuration has codimension $9 - \text{rank } A = 9 - 4 = 5$, with A the matrix whose columns are the exponent vectors of the monomials of G below a row of ones, against 6 independent kinematic invariants in the coefficients [2].

5 Parametric representations

5.1 Schwinger representation

With the Schwinger parameters a_e integrated over $[0, \infty)$ [1],

$$I = \frac{e^{\epsilon\gamma E}}{\Gamma(\nu_1)\Gamma(\nu_2)\Gamma(\nu_3)} \times \int_{\mathbb{R}_+^3} da_1 da_2 da_3 a_1^{\nu_1-1} a_2^{\nu_2-1} a_3^{\nu_3-1} U^{-D/2} e^{-F/U}$$

5.2 Feynman representation

With the same parameters restricted to the simplex $\sum_e a_e = 1$ [1],

$$I = \frac{e^{\epsilon\gamma E}}{\Gamma(\nu_1)\Gamma(\nu_2)\Gamma(\nu_3)} \Gamma\left(-\frac{D}{2} + \nu_1 + \nu_2 + \nu_3\right) \times \int_{\mathbb{R}_+^3} da_1 da_2 da_3 \delta\left(1 - \sum_e a_e\right) a_1^{\nu_1-1} a_2^{\nu_2-1} a_3^{\nu_3-1} F^{D/2-\nu} U^{-D+\nu}$$

5.3 Lee-Pomeransky representation

With the Lee-Pomeransky parameters u_e integrated over $[0, \infty)$ [3],

$$I = \frac{e^{\epsilon\gamma E} \Gamma\left(\frac{D}{2}\right)}{\Gamma(\nu_1)\Gamma(\nu_2)\Gamma(\nu_3)\Gamma(D - \nu_1 - \nu_2 - \nu_3)} \times \int_{\mathbb{R}_+^3} du_1 du_2 du_3 u_1^{\nu_1-1} u_2^{\nu_2-1} u_3^{\nu_3-1} G^{-D/2}$$

Convergence. For Euclidean kinematics, where every coefficient of G has positive real part, and for $\text{Re } D > 0$, the Lee-Pomeransky integral converges absolutely when the real parts of $(\nu_1, \dots, \nu_N)$, divided by $\text{Re}(D/2)$, lie in the interior of the Newton polytope P of G , that is when for every facet $m_j \cdot x \leq b_j$ of P the real part of $b_j D/2 - \sum_e m_{je} \nu_e$ is positive [2]:

$$\begin{aligned} \text{Re}\left(-\frac{D}{2} + \nu_1 + \nu_2 + \nu_3\right) &> 0 \\ \text{Re}(\nu_3) &> 0 \\ \text{Re}(\nu_2) &> 0 \\ \text{Re}(\nu_1) &> 0 \\ \text{Re}(D - \nu_1 - \nu_2 - \nu_3) &> 0 \end{aligned}$$

For such kinematics it converges absolutely for no D and ν_e when P is not full-dimensional.

6 Newton polytope

The Newton polytope P of G is the convex hull of the exponent vectors of its monomials. It has dimension 3 in $\mathbb{R}^3$ and normalised volume 7, with 6 vertices, 9 edges and 5 facets. Its vertices are $v_1 = (1, 0, 0)$, $v_2 = (0, 1, 0)$, $v_3 = (0, 0, 1)$, $v_4 = (2, 0, 0)$, $v_5 = (0, 2, 0)$ and $v_6 = (0, 0, 2)$. Figure 2 draws it.

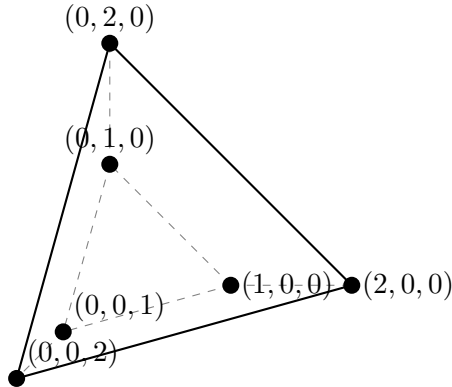


Figure 2: The Newton polytope P of G .

For generic coefficients and non-resonant β the holonomic rank of the GKZ system equals the normalised volume, here 7 [4, 5]. The rank equals the volume for every β exactly when the toric ring $\mathbb{C}[\mathbb{N}A]$ is Cohen-Macaulay [6]. This holds when the graph is one-particle irreducible and one-vertex irreducible, its external momenta are generic enough that no monomial of G cancels, and its propagators are all massive, all massless, or such that every vertex reaches an external leg along massive propagators alone; the configuration is then normal and so Cohen-Macaulay [2, 7, 8]. The number of master integrals is, up to sign, the Euler characteristic of the complement of $\{G = 0\}$ in the torus, at most $N!$ times the Euclidean volume of P , with equality for generic coefficients; graph-polynomial coefficients are rarely generic [9]. The bound equals the normalised volume when the exponent differences span $\mathbb{Z}^N$. feynkit computes neither the holonomic rank at the physical point nor the Euler characteristic, and produces no series solutions, Pfaffian system or restriction to physical kinematics.

7 GKZ system

Writing $G = \sum_j z_j u^{\alpha_j}$ and treating the coefficients z_j as independent variables gives the Euler-Mellin integral

$$I_A(\beta, z) = \int_{\mathbb{R}_+^N} \prod_e du_e u_e^{\nu_e - 1} G(z, u)^{-D/2},$$

the Lee-Pomeransky integral without its prefactor, which the GKZ system annihilates. The system has the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}$$

whose column j is $(1, \alpha_j)$. The parameter vector is $\beta = (-D/2, -\nu_1, \dots, -\nu_N)$, the value the Euler equations require for the Lee-Pomeransky integral [10, 11]. Here

$$\beta = \left(-\frac{D}{2}, -\nu_1, -\nu_2, -\nu_3 \right).$$

The Euler operators are built from $\theta_j = z_j \partial_{z_j}$, which measures the degree in z_j . The operators $E_r = \sum_j A_{rj} \theta_j - \beta_r$ annihilate I_A : row 0 states that I_A is homogeneous of degree $-D/2$ in the z_j , and row i that it has degree $-\nu_i$ under $z_j \mapsto \lambda^{A_{ij}} z_j$, which a rescaling of u_i absorbs:

$$\begin{aligned} E_0 &= \theta_1 + \theta_2 + \theta_3 + \theta_4 + \theta_5 + \theta_6 + \theta_7 + \theta_8 + \theta_9 + \frac{D}{2} \\ E_1 &= 2\theta_1 + \theta_2 + \theta_3 + \theta_7 + \nu_1 \\ E_2 &= \theta_2 + 2\theta_4 + \theta_5 + \theta_8 + \nu_2 \\ E_3 &= \theta_3 + \theta_5 + 2\theta_6 + \theta_9 + \nu_3 \end{aligned}$$

The toric ideal of A is generated by 14 binomials. Each binomial $z^k - z^l$, with $Ak = Al$, gives the operator $\partial^k - \partial^l$; these annihilators of I_A form an analogue of integration-by-parts relations for it [5], and the specialisation to physical coefficients is a separate step. The generators are

$$\begin{aligned} z_5 z_9 - z_6 z_8 \\ z_4 z_6 - z_5^2 \\ z_4 z_9 - z_5 z_8 \\ z_3 z_9 - z_6 z_7 \\ z_2 z_5 - z_3 z_4 \\ z_2 z_6 - z_3 z_5 \\ z_2 z_9 - z_3 z_8 \\ z_2 z_8 - z_4 z_7 \\ z_2 z_9 - z_5 z_7 \\ z_1 z_4 - z_2^2 \\ z_1 z_5 - z_2 z_3 \\ z_1 z_8 - z_2 z_7 \\ z_1 z_6 - z_3^2 \\ z_1 z_9 - z_3 z_7 \end{aligned}$$

8 Symmetries

The group $\text{Aut}(P)$ of unimodular affine maps taking P to itself has order 6 and acts on the vertices listed in Section 6 with orbits $\{v_1, v_2, v_3\}$ and $\{v_4, v_5, v_6\}$. The graph has 6 automorphisms. Of the polytope automorphisms, 1 preserves the coefficients of G . The configuration has 6 symmetry pairs (T, σ) : integer matrices T and permutations σ of the columns of A such that T takes column j of A to column $\sigma(j)$. Each gives the identity $I_A(\beta, z_\sigma) = I_A(T\beta, z)$ for the Euler-Mellin integral I_A of Section 7, with $z_\sigma = (z_{\sigma(1)}, \dots, z_{\sigma(9)})$. Forsgård, Matusevich and Sobieska [12] and de la Cruz [13] print the permutation on the other side, but the substitution in the proof of Corollary 4.1 of [12] gives the form stated here. For I itself the prefactors at β and $T\beta$ enter as well. They are, with σ listed as $(\sigma(1), \sigma(2), \dots)$:

$$\begin{aligned} T_1 &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, & \sigma_1 &= (1, 2, 3, 4, 5, 6, 7, 8, 9) \\ T_2 &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, & \sigma_2 &= (1, 3, 2, 6, 5, 4, 7, 9, 8) \end{aligned}$$

$$\begin{aligned}
T_3 &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, & \sigma_3 &= (4, 2, 5, 1, 3, 6, 8, 7, 9) \\
T_4 &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, & \sigma_4 &= (4, 5, 2, 6, 3, 1, 8, 9, 7) \\
T_5 &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}, & \sigma_5 &= (6, 3, 5, 1, 2, 4, 9, 7, 8) \\
T_6 &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, & \sigma_6 &= (6, 5, 3, 4, 2, 1, 9, 8, 7)
\end{aligned}$$

9 Landau surfaces

The reduced principal A -determinant of G , each irreducible kinematic factor taken once, is the product of the discriminants of G restricted to the faces of P [14, 15, 16]. Each discriminant that is not identically one factorises as follows, by face dimension, with one factor per line; numerical factors and powers of the energy scale μ , which only normalise the coefficients z_j , are left out:

- Dimension 0:

$$\begin{aligned}
&m_1^2 \\
&m_2^2 \\
&m_3^2
\end{aligned}$$

- Dimension 1:

$$\begin{aligned}
&m_1^2 - 2m_1m_2 + m_2^2 - p_1^2 \\
&m_1^2 + 2m_1m_2 + m_2^2 - p_1^2 \\
&m_1^2 - 2m_1m_3 + m_3^2 - p_2^2 \\
&m_1^2 + 2m_1m_3 + m_3^2 - p_2^2 \\
&m_2^2 - 2m_2m_3 + m_3^2 - p_3^2 \\
&m_2^2 + 2m_2m_3 + m_3^2 - p_3^2
\end{aligned}$$

- Dimension 2:

$$\begin{aligned}
&p_1^2 \\
&p_2^2 \\
&p_3^2 \\
&m_1^2 - 2m_1m_2 + m_2^2 - p_1^2 \\
&m_1^2 + 2m_1m_2 + m_2^2 - p_1^2 \\
&m_1^2 - 2m_1m_3 + m_3^2 - p_2^2
\end{aligned}$$

$$\begin{aligned}
& m_1^2 + 2m_1m_3 + m_3^2 - p_2^2 \\
& m_2^2 - 2m_2m_3 + m_3^2 - p_3^2 \\
& m_2^2 + 2m_2m_3 + m_3^2 - p_3^2 \\
& m_1^4p_3^2 + m_1^2m_2^2p_1^2 - m_1^2m_2^2p_2^2 \\
& \quad - m_1^2m_2^2p_3^2 - m_1^2m_3^2p_1^2 + m_1^2m_3^2p_2^2 \\
& \quad - m_1^2m_3^2p_3^2 - m_1^2p_1^2p_3^2 - m_1^2p_2^2p_3^2 \\
& \quad + m_1^2(p_3^2)^2 + m_2^4p_2^2 - m_2^2m_3^2p_1^2 \\
& \quad - m_2^2m_3^2p_2^2 + m_2^2m_3^2p_3^2 - m_2^2p_1^2p_2^2 \\
& \quad + m_2^2(p_2^2)^2 - m_2^2p_2^2p_3^2 + m_3^4p_1^2 \\
& \quad + m_3^2(p_1^2)^2 - m_3^2p_1^2p_2^2 \\
& \quad - m_3^2p_1^2p_3^2 + p_1^2p_2^2p_3^2
\end{aligned}$$

- Dimension 3:

$$\begin{aligned}
& (p_1^2)^2 - 2p_1^2p_2^2 - 2p_1^2p_3^2 \\
& \quad + (p_2^2)^2 - 2p_2^2p_3^2 + (p_3^2)^2 \\
& m_1^4p_3^2 + m_1^2m_2^2p_1^2 - m_1^2m_2^2p_2^2 \\
& \quad - m_1^2m_2^2p_3^2 - m_1^2m_3^2p_1^2 + m_1^2m_3^2p_2^2 \\
& \quad - m_1^2m_3^2p_3^2 - m_1^2p_1^2p_3^2 - m_1^2p_2^2p_3^2 \\
& \quad + m_1^2(p_3^2)^2 + m_2^4p_2^2 - m_2^2m_3^2p_1^2 \\
& \quad - m_2^2m_3^2p_2^2 + m_2^2m_3^2p_3^2 - m_2^2p_1^2p_2^2 \\
& \quad + m_2^2(p_2^2)^2 - m_2^2p_2^2p_3^2 + m_3^4p_1^2 \\
& \quad + m_3^2(p_1^2)^2 - m_3^2p_1^2p_2^2 \\
& \quad - m_3^2p_1^2p_3^2 + p_1^2p_2^2p_3^2
\end{aligned}$$

By comparison with the closed form from the modified Cayley matrix [15], the first-type (Cayley) factors are

$$\begin{aligned}
& m_1 \\
& m_2 \\
& m_3 \\
& m_1^2 - 2m_1m_2 + m_2^2 - p_1^2 \\
& m_1^2 + 2m_1m_2 + m_2^2 - p_1^2 \\
& m_1^2 - 2m_1m_3 + m_3^2 - p_2^2 \\
& m_1^2 + 2m_1m_3 + m_3^2 - p_2^2 \\
& m_2^2 - 2m_2m_3 + m_3^2 - p_3^2 \\
& m_2^2 + 2m_2m_3 + m_3^2 - p_3^2 \\
& m_1^4p_3^2 + m_1^2m_2^2p_1^2 - m_1^2m_2^2p_2^2 \\
& \quad - m_1^2m_2^2p_3^2 - m_1^2m_3^2p_1^2 + m_1^2m_3^2p_2^2 \\
& \quad - m_1^2m_3^2p_3^2 - m_1^2p_1^2p_3^2 - m_1^2p_2^2p_3^2
\end{aligned}$$

$$\begin{aligned}
& + m_1^2 (p_3^2)^2 + m_2^4 p_2^2 - m_2^2 m_3^2 p_1^2 \\
& - m_2^2 m_3^2 p_2^2 + m_2^2 m_3^2 p_3^2 - m_2^2 p_1^2 p_2^2 \\
& + m_2^2 (p_2^2)^2 - m_2^2 p_2^2 p_3^2 + m_3^4 p_1^2 \\
& + m_3^2 (p_1^2)^2 - m_3^2 p_1^2 p_2^2 \\
& - m_3^2 p_1^2 p_3^2 + p_1^2 p_2^2 p_3^2
\end{aligned}$$

and the second-type (Gram) factors are

$$\begin{aligned}
& p_1^2 \\
& p_2^2 \\
& p_3^2 \\
& (p_1^2)^2 - 2p_1^2 p_2^2 - 2p_1^2 p_3^2 \\
& + (p_2^2)^2 - 2p_2^2 p_3^2 + (p_3^2)^2
\end{aligned}$$

The factors are candidate codimension-one singular loci on all sheets of the integral: a point on one of them may or may not be singular on the physical sheet, and the list is not guaranteed complete [16]. The coefficients are specialised to physical kinematics before each face discriminant is computed, so beyond one loop this is the principal Landau determinant rather than the principal A -determinant of the generic polynomial; multiplicities are dropped.

10 Schwinger-representation system

The Schwinger representation gives a second GKZ system on the Cayley configuration of $(\tilde{U}, \tilde{F})$, the Symanzik polynomials with $a_N = 1$ and $a_e = u_e$ otherwise [17, 2]:

$$\begin{aligned}
A_{\text{Cayley}} &= \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 2 & 0 & 1 & 0 \end{bmatrix}, \\
\beta_{\text{Cayley}} &= \left(-D + \nu_1 + \nu_2 + \nu_3, \frac{D}{2} - \nu_1 - \nu_2 - \nu_3, -\nu_1, -\nu_2 \right).
\end{aligned}$$

It is unimodularly equivalent to the Lee-Pomeransky configuration, the matrix A_{LP} and parameter vector $\beta_{\text{LP}} = (-D/2, -\nu_1, \dots, -\nu_N)$ of the GKZ system of G , through

$$T = \begin{bmatrix} 2 & -1 & -1 & -1 \\ -1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix},$$

which maps the parameter vectors as $\beta_{\text{Cayley}} = T\beta_{\text{LP}}$ and the columns of A_{LP} to those of A_{Cayley} up to order; the column correspondence is verified for this integral. Restricting to the $\tilde{F}$ block gives the system

$$\begin{aligned}
A_F &= \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \end{bmatrix}, \\
\beta_F &= \left(\frac{D}{2} - \nu_1 - \nu_2 - \nu_3, -\nu_1, -\nu_2 \right);
\end{aligned}$$

its solutions solve the full system when β_{Cayley} lies in the span of the face's columns, which for the $\tilde{F}$ block means $\nu = (L + 1)D/2$, here $\nu_1 + \nu_2 + \nu_3 = D$ [18]. Away from that value the relation between the two systems is not established.

References

- [1] S. Weinzierl, *Feynman Integrals: A Comprehensive Treatment for Students and Researchers*, Springer, 2022, arXiv:2201.03593.
- [2] R.P. Klausen, *Hypergeometric Feynman Integrals*, PhD thesis, Johannes Gutenberg University Mainz, arXiv:2302.13184 (2023).
- [3] R.N. Lee and A.A. Pomeransky, *Critical points and number of master integrals*, JHEP 11 (2013) 165, arXiv:1308.6676.
- [4] A. Adolphson, *Hypergeometric functions and rings generated by monomials*, Duke Math. J. 73 (1994) 269–290.
- [5] V. Chestnov, F. Gasparotto, M.K. Mandal, P. Mastrolia, S.J. Matsubara-Heo, H.J. Munch and N. Takayama, *Macaulay matrix for Feynman integrals: linear relations and intersection numbers*, JHEP 09 (2022) 187, arXiv:2204.12983.
- [6] L.F. Matusevich, E. Miller and U. Walther, *Homological methods for hypergeometric families*, J. Amer. Math. Soc. 18 (2005) 919, arXiv:math/0406383.
- [7] F. Tellander and M. Helmer, *Cohen-Macaulay property of Feynman integrals*, Commun. Math. Phys. 399 (2023) 1021–1037, arXiv:2108.01410.
- [8] U. Walther, *On Feynman graphs, matroids, and GKZ-systems*, Lett. Math. Phys. 112 (2022) 120, arXiv:2206.05378.
- [9] T. Bitoun, C. Bogner, R.P. Klausen and E. Panzer, *Feynman integral relations from parametric annihilators*, Lett. Math. Phys. 109 (2019) 497–564, arXiv:1712.09215.
- [10] L. de la Cruz, *Feynman integrals as A-hypergeometric functions*, JHEP 12 (2019) 123, arXiv:1907.00507.
- [11] R.P. Klausen, *Hypergeometric series representations of Feynman integrals by GKZ hypergeometric systems*, JHEP 04 (2020) 121, arXiv:1910.08651.
- [12] J. Forsgård, L.F. Matusevich and A. Sobieska, *On transformations of A-hypergeometric functions*, Funkcialaj Ekvacioj 62 (2019) 319, arXiv:1703.03036.
- [13] L. de la Cruz, *Polytope symmetries of Feynman integrals*, Phys. Lett. B 854 (2024) 138744, arXiv:2404.03564.
- [14] I.M. Gelfand, M.M. Kapranov and A.V. Zelevinsky, *Discriminants, Resultants, and Multidimensional Determinants*, Birkhäuser, 1994.
- [15] C. Dlapa, M. Helmer, G. Papathanasiou and F. Tellander, *Symbol alphabets from the Landau singular locus*, JHEP 10 (2023) 161, arXiv:2304.02629.
- [16] C. Fevola, S. Mizera and S. Telen, *Landau singularities revisited: computational algebraic geometry for Feynman integrals*, Phys. Rev. Lett. 132 (2024) 101601, arXiv:2311.14669.
- [17] M. Jimenez-Santacruz, C. Lopez-Arcos and A. Quintero Velez, *Canonical differential equations for Feynman integrals from A-hypergeometric systems in the Schwinger representation*, arXiv:2609.16107.

- [18] R. Britto, T.W. Grimm and A. Hoefnagels, *Resonance and differential reduction of Feynman integrals*, JHEP 09 (2026) 018, arXiv:2606.09978.